

A2RL High-Fidelity Official Simulator



high-fidelity dynamics model provided by A2RL, embeds detailed Dallara EAV24 racecar and Yas Marina Track Model

Vehicle State Feedback (ROS 2)

INS Common Group (100Hz)
IMU Raw Data (250Hz)
GPS Raw Data (100Hz)
Odometry (100Hz)
Opponent states

**Vehicle
Action
Command
Interface**



Low-Level Vehicle Feedback (Virtual CAN)

Wheel speeds
Velocity
Yaw rate
Actuator Status



decision stack runs as external real-time process, all inter-layer timing must survive message transport layer

ROS 2 topic feedback (tactical)

/// /ego_state

/opponent_state
/race_control

(Time-Synchronized Loop)



Virtual-CAN Feedback
(physical)



vCAN
frames

External
Connection

External Real-Time Control Process (Independent Process)

Perception & State Estimation

Planning
(Learned Corridor/SOCP)

Vehicle Control
(MPCC)

CAN Message Abstraction Layer

Real-Time
Constraints
Checkpoints

...

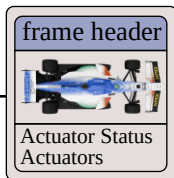
Steer

Throttle

Brake

Gear

Actuation Commands



throttle,
brake,
steering, gear commands